ChatGPT

Computing a_2 from the code

The code defines a computable real a_2 via a series. In fact one finds (after simplifying) that

$$a_2(n) = 2f(n) + \tfrac{1}{2} + 0.25^n = 1 - 2^{-(n+1)} + 2^{-2n} + \epsilon_n,$$

where ϵ_n collects the tiny terms $2 \cdot 72^{-10^{24}} - 2\sum 72^{-10^{24}k}$. Since $72^{-10^{24}}$ is astronomically small, those terms are effectively zero in any practical precision. Using the formula for the sum of a geometric series [1], one checks that

$$\sum_{k=3}^{n+2}(1/2)^k = \frac{(1/2)^3(1-(1/2)^n)}{1-1/2} = \tfrac{1}{4}(1-2^{-n}).$$

Hence $f(n) \approx \frac{1}{4}(1-2^{-n})$ and $a_2(n) \approx 1 - 2^{-n-1} + 2^{-2n}$. In particular for any $n > 1$, $a_2 < 1$ (e.g. $n = 10$ gives $a_2 \approx 0.9995127$). As $n \to \infty$, $a_2 \to 1$. Thus the second therapy rate a_2 is slightly less than 1.

Formulating and solving the optimisation

Jen's total dose constraint is

$$x_1 + a_2\,x_2 = 1,$$

with $x_1, x_2 \geq 0$, and we want to **minimise** total time $T = x_1 + x_2$. Substitute $x_1 = 1 - a_2x_2$ to get

$$T = 1 - a_2x_2 + x_2 \;=\; 1 + (1-a_2)x_2.$$

Since $a_2 < 1$, the coefficient $1 - a_2 > 0$, so T increases with x_2. Thus the minimum is obtained by making x_2 as small as possible, namely $x_2 = 0$ and $x_1 = 1$. Equivalently, geometrically this is a linear objective over a line segment, whose minimum occurs at an endpoint (vertex) [2]. (If $a_2 > 1$ the opposite end would be chosen, but here $a_2 < 1$.)

Hence the optimal solution is

$$x_1 = 1, \quad x_2 = 0.$$

In two-decimal form, to ℓ^∞–norm accuracy ± 0.005, this is $x_1 = 1.00$, $x_2 = 0.00$. That is, Jen should receive the full dose from the first (faster) therapy; the second therapy is slower ($a_2 < 1$) and so it is optimal to set $x_2 = 0$.

Answer: $x_1 \approx 1.00,\ x_2 \approx 0.00.$

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[1] 7.3 - Geometric Sequences
https://people.richland.edu/james/lecture/m116/sequences/geometric.html

2 Sparse Convex Optimization Methods for Machine Learning
https://www.research-collection.ethz.ch/bitstreams/8071406e-89fe-4550-8c53-0cce92cebb38/download